

ANSWERS

1.

- (a) (region of space) where a mass experiences a force B1
- (b) (i) potential energy = $(-)\frac{GMm}{x}$ C1
 $\Delta E_p = \frac{GMm}{2R} - \frac{GMm}{3R}$ M1
 $= \frac{GMm}{6R}$ A0
- (ii) $E_k = \frac{1}{2}m(7600^2 - 7320^2)$ M1
 $= (2.09 \times 10^6)m$ A0
- (c) (i) $2.09 \times 10^6 = (6.67 \times 10^{-11} M)/(6 \times 3.4 \times 10^6)$ C1
 $M = 6.39 \times 10^{23} \text{ kg}$ A1
- (ii) e.g. no energy dissipated due to friction with atmosphere/air
 rocket is outside atmosphere
 not influenced by another planet etc. B1

2.

- (a) force per unit mass (*ratio idea essential*) B1
- (b) $g = \frac{GM}{R^2}$ C1
 $8.6 \times (0.6 \times 10^7)^2 = M \times 6.67 \times 10^{-11}$ C1
 $M = 4.6 \times 10^{24} \text{ kg}$ A1
- (c) (i) *either* potential decreases as distance from planet decreases
or potential zero at infinity and X is closer to zero
or potential $\propto -1/r$ and Y more negative M1
 so point Y is closer to planet. A1
- (ii) idea of $\Delta\phi = \frac{1}{2}v^2$ C1
 $(6.8 - 5.3) \times 10^7 = \frac{1}{2}v^2$
 $v = 5.5 \times 10^3 \text{ ms}^{-1}$ A1

3.

(a) (i) force per (unit) mass(ratio idea essential) B1

(ii) $g = GM / R^2$ C1

$9.81 = (6.67 \times 10^{-11} \times M) / (6.38 \times 10^6)^2$ (all 3 s.f) M1

$M = 5.99 \times 10^{24} \text{ kg}$ A0

(b) (i) either $GM = \omega^2 r^3$ or $gR^2 = \omega^2 r^3$ C1

either $6.67 \times 10^{-11} \times 5.99 \times 10^{24} = \omega^2 \times (2.86 \times 10^7)^3$

or $9.81 \times (6.38 \times 10^6)^2 = \omega^2 \times (2.86 \times 10^7)^3$ C1

$\omega = 1.3 \times 10^{-4} \text{ rad s}^{-1}$ A1

(use of $r = 2.22 \times 10^7 \text{ m}$ scores max 2 marks)

(ii) period of orbit = $2\pi / \omega$ C1

= $4.8 \times 10^4 \text{ s}$ (= 13.4 hours) A1

period for geostationary satellite is 24 hours (= $8.6 \times 10^4 \text{ s}$) A1

so no A0

(c) satellite can then provide cover at Poles B1

4.

(a) $F \propto Mm / R^2$ (words or explained symbols) M1

either M and m are point masses

or $R \gg$ diameter of masses ... (do not allow 'size') A1

(b) (i) equatorial orbit B1

period 24 hours / same angular speed B1

from west to east / same direction of rotation B1

(allow one of the last two marks for 'always overhead' if 2nd or 3rd marks not scored)

(ii) gravitational force provides centripetal force

/ gives rise to centripetal acceleration(in 'words') B1

$GM / x^2 = x\omega^2$ M1

$g = GM / R^2$ M1

to give $gR^2 = x^3 \omega^2$ A0

(iii) $\omega = 2\pi / (24 \times 3600) = 7.27 \times 10^{-5} \text{ rad s}^{-1}$ C1

$9.81 \times (6.4 \times 10^6)^2 = x^3 \times (7.27 \times 10^{-5})^2$ C1

$x^3 = 7.6 \times 10^{22}$

$x = 4.2 \times 10^7 \text{ m}$ A1

(use of $g = 10 \text{ m s}^{-2}$, loses 1 mark but once only in the Paper)

5.		
(a)	work done moving <u>unit</u> mass from infinity to the point	M1 A1
(b) (i)	at R , $\phi = 6.3 \times 10^7 \text{ J kg}^{-1}$ (allow $\pm 0.1 \times 10^7$) $\phi = GM / R$ $6.3 \times 10^7 = (6.67 \times 10^{-11} \times M) / (6.4 \times 10^6)$ $M = 6.0 \times 10^{24} \text{ kg}$ (allow $5.95 \rightarrow 6.14$) Maximum of 2/3 for any value chosen for ϕ not at R	B1 C1 A1
(ii)	change in potential = $2.1 \times 10^7 \text{ J kg}^{-1}$ (allow $\pm 0.1 \times 10^7$) loss in potential energy = gain in kinetic energy $\frac{1}{2} mv^2 = \phi m$ or $\frac{1}{2} mv^2 = GM / 3R$ $\frac{1}{2} v^2 = 2.1 \times 10^7$ $v = 6.5 \times 10^3 \text{ m s}^{-1}$(allow $6.3 \rightarrow 6.6$) (answer $7.9 \times 10^3 \text{ m s}^{-1}$, based on $x = 2R$, allow max 3 marks)	C1 B1 C1 A1
(iii)	e.g. speed / velocity / acceleration would be greater deviates / bends from straight path (any sensible ideas, 1 each, max 2)	B1 B1
6.		
(a)	force per unit mass (<i>ratio idea essential</i>)	B1
(b)	graph: correct curvature from $(R, 1.0 g_s)$ & at least one other correct point	M1 A1
(c) (i)	fields of Earth and Moon are in opposite directions <i>either</i> resultant field found by subtraction of the field strength <i>or</i> any other sensible comment so there is a point where it is zero (allow $F_E = -F_M$ for 2 marks)	M1 A1 A0
(ii)	$GM_E / x^2 = GM_M / (D - x)^2$ $(6.0 \times 10^{24}) / (7.4 \times 10^{22}) = x^2 / (60R_E - x)^2$ $x = 54 R_E$	C1 C1 A1
(iii)	graph: $g = 0$ at least $\frac{2}{3}$ distance to Moon g_E and g_M in opposite directions correct curvature (by eye) and $g_E > g_M$ at surface	B1 M1 A1

7.

- (a) (i) force proportional to product of masses B1
 force inversely proportional to square of separation B1
- (ii) separation much greater than radius / diameter of Sun / planet B1
- (b) (i) e.g. force or field strength $\propto 1/r^2$
 potential $\propto 1/r$ B1
- (ii) e.g. gravitational force (always) attractive B1
 electric force attractive or repulsive B1

8.

- (a) work done in bringing unit mass from infinity (to the point) B1
- (b) gravitational force is (always) attractive B1
either as r decreases, object/mass/body does work
or work is done by masses as they come together B1
- (c) *either* force on mass = mg (where g is the acceleration of free fall /gravitational field strength) B1
 $g = GM/r^2$ B1
 if $r \approx h$, g is constant B1
 $\Delta E_p = \text{force} \times \text{distance moved}$ M1
 $= mgh$ A0
or $\Delta E_p = m\Delta\phi$ (C1
 $= GMm(1/r_1 - 1/r_2) = GMm(r_2 - r_1)/r_1r_2$ (B1
 if $r_2 \approx r_1$, then $(r_2 - r_1) = h$ and $r_1r_2 = r^2$ (B1
 $g = GM/r^2$ (B1
 $\Delta E_p = mgh$ (A0
- (d) $\frac{1}{2}mv^2 = m\Delta\phi$
 $v^2 = 2 \times GM/r$ C1
 $= (2 \times 4.3 \times 10^{13}) / (3.4 \times 10^6)$ C1
 $v = 5.0 \times 10^3 \text{ m s}^{-1}$ A1
 (Use of diameter instead of radius to give $v = 3.6 \times 10^3 \text{ m s}^{-1}$ scores 2 marks)

9.

- (a) force proportional to product of masses and inversely proportional to
square of separation (*do not allow square of distance/radius*) M1
either point masses *or* separation @ size of masses A1
- (b) (i) $\omega = 2\pi / (27.3 \times 24 \times 3600)$ or $2\pi / (2.36 \times 10^6)$ M1
 $= 2.66 \times 10^{-6} \text{ rad s}^{-1}$ A0
- (ii) $GM = r^3 \omega^2$ or $GM = v^2 r$ C1
 $M = (3.84 \times 10^5 \times 10^3)^3 \times (2.66 \times 10^{-6})^2 / (6.67 \times 10^{-11})$ M1
 $= 6.0 \times 10^{24} \text{ kg}$ A0
(special case: uses $g = GM/r^2$ with $g = 9.81$, $r = 6.4 \times 10^6$ scores max 1 mark
- (c) (i) grav. force $= (6.0 \times 10^{24}) \times (7.4 \times 10^{22}) \times (6.67 \times 10^{-11}) / (3.84 \times 10^8)^2$ C1
 $= 2.0 \times 10^{20} \text{ N}$ (*allow 1 SF*) A1
- (ii) *either* $\Delta E_p = Fx$ because F constant as $x \ll$ radius of orbit B1
 $\Delta E_p = 2.0 \times 10^{20} \times 4.0 \times 10^{-2}$ C1
 $= 8.0 \times 10^{18} \text{ J}$ (*allow 1 SF*) A1
- or* $\Delta E_p = GMm/r_1 - GMm/r_2$ C1
Correct substitution B1
 $8.0 \times 10^{18} \text{ J}$ A1
($\Delta E_p = GMm/r_1 + GMm/r_2$ is incorrect physics so 0/3)

10.

- (a) force is proportional to the product of the masses and
inversely proportional to the square of the separation M1
either point masses *or* separation $\gg$ size of masses A1
- (b) (i) gravitational force provides the centripetal force B1
 $mv^2/r = GMm/r^2$ and $E_k = \frac{1}{2}mv^2$ M1
hence $E_k = GMm/2r$ A0
- (ii) 1. $\Delta E_k = \frac{1}{2} \times 4.00 \times 10^{14} \times 620 \times (\{7.30 \times 10^6\}^{-1} - \{7.34 \times 10^6\}^{-1})$ C1
 $= 9.26 \times 10^7 \text{ J}$ (*ignore any sign in answer*) A1
(*allow* $1.0 \times 10^8 \text{ J}$ if evidence that E_k evaluated separately for each r)
2. $\Delta E_p = 4.00 \times 10^{14} \times 620 \times (\{7.30 \times 10^6\}^{-1} - \{7.34 \times 10^6\}^{-1})$ C1
 $= 1.85 \times 10^8 \text{ J}$ (*ignore any sign in answer*) A1
(*allow* 1.8 or $1.9 \times 10^8 \text{ J}$)
- (iii) *either* $(7.30 \times 10^6)^{-1} - (7.34 \times 10^6)^{-1}$ or ΔE_k is positive / E_k increased M1
speed has increased A1

11.

- (a) region of space area / volume B1
 where a mass experiences a force B1
- (b) (i) force proportional to product of two masses M1
 force inversely proportional to the square of their separation M1
either reference to point masses *or* separation >> 'size' of masses A1
- (ii) field strength = GM / x^2 **or** field strength $\propto 1 / x^2$ C1
 ratio = $(7.78 \times 10^8)^2 / (1.5 \times 10^8)^2$ C1
 = 27 A1
- (c) (i) *either* centripetal force = $mR\omega^2$ and $\omega = 2\pi / T$
or centripetal force = mv^2 / R and $v = 2\pi R / T$ B1
 gravitational force provides the centripetal force B1
either $GMm / R^2 = mR\omega^2$ *or* $GMm / R^2 = mv^2 / R$ M1
 $M = 4\pi^2 R^3 / GT^2$ A0
 (allow working to be given in terms of acceleration)
- (ii) $M = \{4\pi^2 \times (1.5 \times 10^{11})^3\} / \{6.67 \times 10^{-11} \times (3.16 \times 10^7)^2\}$ C1
 = 2.0×10^{30} kg A1

12.

- (a) equatorial orbit / above equator B1
 satellite moves from west to east / same direction as Earth spins B1
 period is 24 hours / same period as spinning of Earth B1
 (allow 1 mark for 'appears to be stationary/overhead' if none of above marks scored)
- (b) gravitational force provides/is the centripetal force B1
 $GMm/R^2 = mR\omega^2$ *or* $GMm/R^2 = mv^2/R$ M1
 $\omega = 2\pi / T$ *or* $v = 2\pi R / T$ *or* clear substitution M1
 clear working to give $R^3 = (GMT^2 / 4\pi^2)$ A1
- (c) $R^3 = 6.67 \times 10^{-11} \times 6.0 \times 10^{24} \times (24 \times 3600)^2 / 4\pi^2$ C1
 = 7.57×10^{22} C1
 $R = 4.2 \times 10^7$ m A1
 (missing out 3600 gives 1.8×10^5 m and scores 2/3 marks)

13.

(a) work done in moving unit mass from infinity (to the point) M1
A1

(b) (i) gravitational potential energy = GMm / x
energy = $(6.67 \times 10^{-11} \times 7.35 \times 10^{22} \times 4.5) / (1.74 \times 10^6)$ M1
energy = 1.27×10^7 J A0

(ii) change in grav. potential energy = change in kinetic energy B1
 $\frac{1}{2} \times 4.5 \times v^2 = 1.27 \times 10^7$
 $v = 2.4 \times 10^3 \text{ ms}^{-1}$ A1

(c) Earth would attract the rock / potential at Earth('s surface) not zero / <0 M1
/ at Earth, potential due to Moon not zero A1
escape speed would be lower

14.

(a) force proportional to product of the two masses and inversely proportional to the square of their separation M1
either reference to point masses *or* separation >> 'size' of masses A1

(b) gravitational force provides the centripetal force B1
 $GMm / R^2 = mR\omega^2$ M1
where m is the mass of the planet A1
 $GM = R^3\omega^2$ A0

(c) $\omega = 2\pi / T$ C1
either $M_{\text{star}} / M_{\text{Sun}} = (R_{\text{star}} / R_{\text{Sun}})^3 \times (T_{\text{Sun}} / T_{\text{star}})^2$
 $M_{\text{star}} = 4^3 \times (1/2)^2 \times 2.0 \times 10^{30}$ C1
 $= 3.2 \times 10^{31} \text{ kg}$ A1
or $M_{\text{star}} = (2\pi)^2 R_{\text{star}}^3 / GT^2$ (C1)
 $= \{(2\pi)^2 \times (6.0 \times 10^{11})^3\} / \{6.67 \times 10^{-11} \times (2 \times 365 \times 24 \times 3600)^2\}$ (C1)
 $= 3.2 \times 10^{31} \text{ kg}$ (A1)

15.

- (a) work done bringing unit mass
from infinity (to the point)

M1
A1

(b) $E_p = -m\phi$

B1

(c) $\phi \propto 1/x$

C1

either at $6R$ from centre, potential is $(6.3 \times 10^7)/6$ ($= 1.05 \times 10^7 \text{ J kg}^{-1}$)
and at $5R$ from centre, potential is $(6.3 \times 10^7)/5$ ($= 1.26 \times 10^7 \text{ J kg}^{-1}$)
change in energy $= (1.26 - 1.05) \times 10^7 \times 1.3$
 $= 2.7 \times 10^6 \text{ J}$

C1

C1

A1

or change in potential $= (1/5 - 1/6) \times (6.3 \times 10^7)$
change in energy $= (1/5 - 1/6) \times (6.3 \times 10^7) \times 1.3$
 $= 2.7 \times 10^6 \text{ J}$

(C1)

(C1)

(A1)

16.

- (a) gravitational force provides/is the centripetal force

B1

$$GMm/r^2 = mv^2/r$$

M1

$$v = \sqrt{GM/r}$$

A0

allow gravitational field strength provides/is the centripetal acceleration

(B1)

$$GM/r^2 = v^2/r$$

(M1)

- (b) (i) kinetic energy increase/change = loss/change in (gravitational) potential energy

B1

$$\frac{1}{2}mV_0^2 = GMm/x$$

C1

$$V_0^2 = 2GM/x$$

$$V_0 = \sqrt{2GM/x}$$

A1

(max. 2 for use of r not x)

- (ii) V_0 is (always) greater than v (for $x = r$)
so stone could not enter into orbit

M1

A1

(expressions in (a) and (b)(i) must be dimensionally correct)

17.

(a) $g = GM/R^2$ C1
 $= (6.67 \times 10^{-11} \times 6.4 \times 10^{23}) / (3.4 \times 10^6)^2 = 3.7 \text{ N kg}^{-1}$ A1

(b) $\Delta E_p = mg\Delta h$
 because $\Delta h \ll R$ (or $1800 \text{ m} \ll 3.4 \times 10^6 \text{ m}$) g is constant B1
 $\Delta E_p = 2.4 \times 3.7 \times 1800$ C1
 $= 1.6 \times 10^4 \text{ J}$ A1
 (use of $g = 9.8 \text{ m s}^{-2}$ max. 1 for explanation)

(c) gravitational potential energy $= (-)GMm/x$ C1
 $v^2 = 2GM/x$ C1
 $x = 4D = 4 \times 6.8 \times 10^6$ C1
 $v^2 = (2 \times 6.67 \times 10^{-11} \times 6.4 \times 10^{23}) / (4 \times 6.8 \times 10^6)$
 $= 3.14 \times 10^6$
 $v = 1.8 \times 10^3 \text{ m s}^{-1}$ A1
 (use of 3.5D giving $1.9 \times 10^3 \text{ m s}^{-1}$, allow max. 3)

18.

(a) (gravitational) force proportional to product of masses and inversely proportional to square of separation M1
 reference to *either* point masses *or* particles *or* 'size' much less than separation A1

(b) gravitational force provides/is the centripetal force B1
 $GM_N m / r^2 = m r \omega^2$ (or $m v^2 / r$) M1
 $2\pi / T$ (or $v = 2\pi r / T$) leading to $GM_N = 4\pi^2 r^3 / T^2$ A1

(c) $M_N / M_U = (3.55/5.83)^3 \times (13.5/5.9)^2$
 x^3 factor correct C1
 T^2 factor correct C1
 ratio = 1.18 (allow 1.2) A1

alternative method: mass of Neptune = $1.019 \times 10^{26} \text{ kg}$ (C1)
 mass of Uranus = $8.621 \times 10^{25} \text{ kg}$ (C1)
 ratio = 1.18 (A1)

19.

- (a) (i) 1. $F = Gm_1m_2/x^2$
 $= (6.67 \times 10^{-11} \times 2.50 \times 5.98 \times 10^{24}) / (6.37 \times 10^6)^2$ M1
 $= 24.6 \text{ N (accept 2 s.f. or more)}$ A1
2. $F = mx\omega^2$ or $F = mv^2/x$ and $v = \omega x$ (accept x or r for distance) C1
 $= 2.50 \times 6.37 \times 10^6 \times (2\pi/24 \times 3600)^2$
 $= 0.0842 \text{ N (accept 2 s.f. or more)}$ A1
- (ii) reading $= 24.575 - 0.0842$ B1
 $= 24.5 \text{ N (accept only 3 s.f.)}$ A1
- (b) gravitational force provides the centripetal force M1
gravitational force is 'equal' to the centripetal force
(accept $Gm_1m_2/x^2 = mx\omega^2$ or $F_C = F_G$) M1
'weight'/sensation of weight/contact force/reaction force is difference between F_G
and F_C which is zero A1

20.

- (a) (i) gravitational force provides/is the centripetal force B1
- $GMm_S/x^2 = m_Sv^2/x$ (allow x or r , allow m or m_S) M1
- $E_K = \frac{1}{2}m_Sv^2$ and clear algebra leading to $E_K = GMm_S/2x$ A1
- (ii) $E_P = -GMm_S/x$ (sign essential) B1
- (iii) $E_T = E_K + E_P$
 $= GMm_S/2x - GMm_S/x$ C1
 $= -GMm_S/2x$ (allow ECF from (a)(ii)) A1
- (b) (i) decreases B1
- (ii) decreases B1
- (iii) decreases B1
- (iv) increases B1
- (for answers in (b) allow ECF from (a)(iii))

21.

(a) (gravitational) force proportional to product of masses
and inversely proportional to square of separation M1
either point masses *or* particles *or* 'size' $\ll$ separation A1

(b) gravitational force provides the centripetal force B1

either $GMm/x^2 = mx\omega^2$ *or* mv^2/x M1

either $\omega = 2\pi/T$ *or* $v = 2\pi x/T$ and working to $GM = 4\pi^2 x^3/T^2$ A1

(c) *either* use of gradient of graph *or* line through origin so can use single point
or line shown extrapolated to origin B1

gradient = $(4.5 \times 10^{14})/0.35$

$6.67 \times 10^{-11} \times M = 4\pi^2 \times (4.5 \times 10^{14} \times 10^9)/(0.35 \times \{24 \times 3600\}^2)$

correct conversion for km^3 and power of 10 C1

correct conversion for day^2 C1

$M = 1.02 \times 10^{26} \text{ kg}$ A1

22.

(a) force proportional to product of the (two) masses and inversely
proportional to the square of their separation M1
either reference to point masses *or* separation $\ll$ 'size' of masses A1

(b) gravitational force provides/is the centripetal force B1

$GMm/r^2 = mv^2/r$ *or* $GMm/r^2 = mr\omega^2$ and $v = r\omega$
and algebra leading to $v = (GM/r)^{1/2}$

B1

(c) (i) 1. $v_A/v_B = (r_B/r_A)^{1/2}$
 $= (2.2 \times 10^{10}/1.3 \times 10^8)^{1/2}$ C1
 $= 13 \text{ (13.0)}$ A1

2. $v = 2\pi r/T$ *or* $v \propto r/T$ *or* $vT/r = \text{constant}$ C1

$T_A/T_B = (r_A/r_B) \times (v_B/v_A)$
 $= (1.3 \times 10^8/2.2 \times 10^{10}) \times (1/13)$ C1
 $= 4.5 \text{ (4.54)} \times 10^{-4}$ A1

or

$$T^2 = 4\pi^2 r^3 / GM \text{ or } T^2 \propto r^3 \text{ or } T^2 / r^3 = \text{constant} \quad (\text{C1})$$

$$\begin{aligned} T_A / T_B &= (r_A^3 / r_B^3)^{1/2} \\ &= [(1.3 \times 10^8)^3 / (2.2 \times 10^{10})^3]^{1/2} \quad (\text{C1}) \\ &= 4.5 \text{ (4.54)} \times 10^{-4} \quad (\text{A1}) \end{aligned}$$

(ii) $T = 2\pi / 1.7 \times 10^{-4}$
 $= 3.70 \times 10^4 \text{ s} \quad \text{C1}$
 $T_B = 3.70 \times 10^4 / 4.54 \times 10^{-4}$
 $= 8.1 \times 10^7 \text{ s} \quad \text{A1}$
 If identifies T_A as T_B then 0/2

23.

(a) (gravitational) potential at infinity defined as/is zero B1

(gravitational) force attractive so work got out/done as object moves from infinity
 (so potential is negative) B1

(b) (i) $\Delta E = m\Delta\phi$
 $= 180 \times (14 - 10) \times 10^8 \quad \text{C1}$
 $= 7.2 \times 10^{10} \text{ J} \quad \text{A1}$

increase B1

(ii) energy required $= 180 \times (10 - 4.4) \times 10^8$
 or
 energy per unit mass $= (10 - 4.4) \times 10^8 \quad \text{C1}$
 $\frac{1}{2} \times 180 \times v^2 = 180 \times (10 - 4.4) \times 10^8$
 or
 $\frac{1}{2} \times v^2 = (10 - 4.4) \times 10^8 \quad \text{C1}$
 $v = 3.3 \times 10^4 \text{ ms}^{-1} \quad \text{A1}$

24.

(a) (i) gravitational force provides/is the centripetal force B1

same gravitational force (by Newton III) B1

(ii) $\omega = 2\pi / T$
 $= 2\pi / (4.0 \times 365 \times 24 \times 3600) \quad \text{C1}$
 $= 5.0 \text{ (4.98)} \times 10^{-8} \text{ rad s}^{-1} \quad \text{A1}$

- (b) (i) (centripetal force =) $M_A d \omega^2 = M_B (2.8 \times 10^8 - d) \omega^2$
or
 $M_A d_A = M_B d_B$ C1
- $M_A / M_B = 3.0 = (2.8 \times 10^8 - d) / d$ C1
- $d = 7.0 \times 10^7 \text{ km}$ A1
- (ii) $GM_A M_B / (2.8 \times 10^{11})^2 = M_A d \omega^2$ B1
- $M_B = (2.8 \times 10^{11})^2 \times d \omega^2 / G$
 $= (2.8 \times 10^{11})^2 \times (7.0 \times 10^{10}) \times (4.98 \times 10^{-8})^2 / (6.67 \times 10^{-11})$ C1
- $= 2.0 \times 10^{29} \text{ kg}$ A1

25.

- (a) gravitational force provides/is the centripetal force B1
- $GMm/r^2 = mv^2/r$ or $GMm/r^2 = mr\omega^2$
and $v = 2\pi r / T$ or $\omega = 2\pi / T$ M1
- with algebra to $T^2 = 4\pi^2 r^3 / GM$ A1
- or
- acceleration due to gravity is the centripetal acceleration (B1)
- $GM/r^2 = v^2/r$ or $GM/r^2 = r\omega^2$
and $v = 2\pi r / T$ or $\omega = 2\pi / T$ (M1)
- with algebra to $T^2 = 4\pi^2 r^3 / GM$ (A1)
- (b) (i) equatorial orbit/orbits (directly) above the equator B1
- from west to east B1
- (ii) $(24 \times 3600)^2 = 4\pi^2 r^3 / (6.67 \times 10^{-11} \times 6.0 \times 10^{24})$ C1
- $r^3 = 7.57 \times 10^{22}$
- $r = 4.2 \times 10^7 \text{ m}$ A1

$$(c) \quad (T/24)^2 = \{(2.64 \times 10^7)/(4.23 \times 10^7)\}^3 \quad \text{B1}$$

$$= 0.243$$

$$T = 12 \text{ hours} \quad \text{A1}$$

or

$$k (= T^2/r^3) = 24^2/(4.23 \times 10^7)^3 \quad \text{(B1)}$$

$$= 7.61 \times 10^{-21}$$

$$T^2 (= kr^3) = 7.61 \times 10^{-21} \times (2.64 \times 10^7)^3$$

$$= 140$$

$$T = 12 \text{ hours} \quad \text{(A1)}$$

26.

(a) force per unit mass B1

(b) (i) radius/diameter/size (of Proxima Centauri) << /is much less than
 4.0×10^{13} km/separation (of Sun and star)

or

(because) it is a uniform sphere B1

(ii) 1. field strength = GM/x^2

$$= (6.67 \times 10^{-11} \times 2.5 \times 10^{29})/(4.0 \times 10^{13} \times 10^3)^2 \quad \text{C1}$$

$$= 1.0 \times 10^{-14} \text{ N kg}^{-1} \quad \text{A1}$$

2. force = field strength $\times$ mass

$$= 1.0 \times 10^{-14} \times 2.0 \times 10^{30} \quad \text{C1}$$

or

$$\text{force} = GMm/x^2$$

$$= (6.67 \times 10^{-11} \times 2.5 \times 10^{29} \times 2.0 \times 10^{30})/(4.0 \times 10^{13} \times 10^3)^2 \quad \text{(C1)}$$

$$= 2.0 \times 10^{16} \text{ N} \quad \text{A1}$$

(c) force (of 2×10^{16} N) would have little effect on (large) mass of Sun B1

would cause an acceleration of Sun of $1.0 \times 10^{-14} \text{ ms}^{-2}$ /very small/negligible acceleration B1

or

many stars all around the Sun (B1)

net effect of forces/fields is zero (B1)

27.

1(a)	gravitational force (of attraction between satellite and planet)	B1
	<u>provides / is</u> centripetal force (on satellite about the planet)	B1
1(b)	$M = (4/3) \times \pi R^3 \rho$	B1
	$\omega = 2\pi / T$ or $v = 2\pi nR / T$	B1
	$GM / (nR)^2 = nR\omega^2$ or v^2 / nR	M1
	substitution clear to give $\rho = 3\pi n^3 / GT^2$	A1
1(c)	$n = (3.84 \times 10^5) / (6.38 \times 10^3) = 60.19$ or 60.2	C1
	$\rho = 3\pi \times 60.19^3 / [(6.67 \times 10^{-11}) \times (27.3 \times 24 \times 3600)^2]$	C1
	$\rho = 5.54 \times 10^3 \text{ kg m}^{-3}$	A1

28.

1(a)	force per unit mass	B1
1(b)(i)	$g = GM / r^2$	C1
	$= (6.67 \times 10^{-11} \times 1.0 \times 10^{13}) / (3.6 \times 10^3)^2$	
	$= 5.1 \times 10^{-5} \text{ N kg}^{-1}$	A1
1(b)(ii)	mass = (960 / 9.81) kg	C1
	weight on comet = (960 / 9.81) $\times 5.1 \times 10^{-5}$	
	$= 5.0 \times 10^{-3} \text{ N}$	A1
1(c)	similarity: e.g. both attractive/pointed towards the comet e.g. same order of magnitude	B1
	difference: e.g. radial/non-radial e.g. same (over surface)/varies (over surface)	B1

29.

3(a)	force per unit mass	B1
3(b)	changes in height <u>much</u> less than radius of Earth	M1
	so (radial) field lines are almost parallel or	A1
	$g = GM / R^2 \approx GM / (R + h)^2$	

3(c)	gravitational force provides/is centripetal force	B1
	$G M m / r^2 = m v^2 / r$	C1
	$v = (2\pi \times 1.5 \times 10^{11}) / (3600 \times 24 \times 365) = 2.99 \times 10^4 \text{ (ms}^{-1}\text{)}$	C1
	$6.67 \times 10^{-11} M = 1.5 \times 10^{11} \times (2.99 \times 10^4)^2$	C1
	$M = 2.0 \times 10^{30} \text{ kg}$	A1
	or	
	$G M m / r^2 = m r \omega^2$	(C1)
	$\omega = 2\pi / (3600 \times 24 \times 365) = 1.99 \times 10^{-7} \text{ (rad s}^{-1}\text{)}$	(C1)
	$6.67 \times 10^{-11} M = (1.5 \times 10^{11})^3 \times (1.99 \times 10^{-7})^2$	(C1)
	$M = 2.0 \times 10^{30} \text{ kg}$	(A1)
	or	
	$T^2 = 4\pi^2 r^3 / G M$	(C2)
	$M = 4\pi^2 \times (1.5 \times 10^{11})^3 / ((3600 \times 24 \times 365)^2 \times 6.67 \times 10^{-11})$	(C1)
	$= 2.0 \times 10^{30} \text{ kg}$	(A1)

30.

1(a)	force proportional to <u>product</u> of masses and inversely proportional to square of separation	B1
	idea of <u>force</u> between <u>point</u> masses	B1
1(b)	mass of Jupiter (M) = $(4/3)\pi R^3 \rho$	B1
	$\omega = 2\pi / T$ or $v = 2\pi n R / T$	B1
	$(m)\omega^2 x = G M(m) / x^2$ or $(m)v^2 / x = G M(m) / x^2$	M1
	substitution and correct algebra leading to $\rho T^2 = 3\pi n^3 / G$	A1
1(c)(i)	$n = (4.32 \times 10^5) / (7.15 \times 10^4)$ or $n = 6.04$	C1
	$\rho \times (42.5 \times 3600)^2 = (3\pi \times 6.04^3) / (6.67 \times 10^{-11})$	C1
	$\rho = 1.33 \times 10^3 \text{ kg m}^{-3}$	A1
1(c)(ii)	Jupiter likely to be a gas/liquid (at high pressure) [allow other sensible suggestions]	B1

31.

1(a)(i)	either direction of force on a (small test) mass or direction of acceleration of a (small test) mass	B1
1(a)(ii)	Any three from: - the lines are radial - near the surface the lines are (approximately) parallel - parallel lines so constant field strength - constant field strength hence constant acceleration of free fall	B3
1(b)(i)	$g = GM/R^2$ $g = (6.67 \times 10^{-11} \times 7.35 \times 10^{22}) / (1.74 \times 10^3 \times 10^3)^2$ $g = 1.62 \text{ N kg}^{-1}$	C1 A1
1(b)(ii)	either $x\omega^2 = GM/x^2$ and $\omega = 2\pi/T$ or $v^2/x = GM/x^2$ and $v = 2\pi r/T$ $(1.74 \times 10^6 + 320 \times 10^3)^3 \times 4\pi^2 / T^2 = (6.67 \times 10^{-11} \times 7.35 \times 10^{22})$ $T^2 = 7.04 \times 10^7$ $T = 8400 \text{ s (8390)}$	C1 C1 A1

32.

1(a)(i)	direction of force on a (small test) <u>mass</u> or path in which a (small test) <u>mass</u> will move	B1
1(a)(ii)	(at surface,) lines (of force) are radial	B1
	Earth has large radius/height above surface is small so lines are (approximately) parallel	B1
	parallel lines → constant field strength	B1
1(b)	(change in) KE of rock = (change in) PE or $\frac{1}{2}mv^2 = GMm/R$ $(m)v^2 = (m)(2 \times 6.67 \times 10^{-11} \times 7.4 \times 10^{22}) / (1.7 \times 10^3 \times 10^3)$ $v = 2.4 \times 10^3 \text{ m s}^{-1}$ correct conclusion based on <u>comparison</u> of v with 2.8 km s^{-1} or (change in) KE of rock = (change in) PE (at infinity) $E_p = (6.67 \times 10^{-11} \times 7.4 \times 10^{22} \times m) / (1.7 \times 10^3 \times 10^3)$ $= 2.9 \times 10^6 m$ $E_K \text{ of rock} = \frac{1}{2} \times m \times (2.8 \times 10^3)^2 = 3.9 \times 10^6 m$ correct conclusion based on <u>comparison</u> of E_K and E_p values or	C1 C1 A1 B1 (C1) (C1) (A1) (B1)

	(change in) KE of rock = (change in) PE or $\frac{1}{2}mv^2 = GMm/R$	(C1)
	$(m) (2800)^2 = (m) (2 \times 6.67 \times 10^{-11} \times 7.4 \times 10^{22}) / R$	(C1)
	$R = 1.3 \times 10^3 \text{ km}$	(A1)
	correct conclusion based on <u>comparison</u> of R with $1.7 \times 10^3 \text{ km}$	(B1)
	or	
	(change in) KE of rock = (change in) PE or $\frac{1}{2}mv^2 = GMm/R$	(C1)
	$(m) (2800)^2 = (m) (2 \times 6.67 \times 10^{-11} \times M) / (1.7 \times 10^6)$	(C1)
	$M = 1.0 \times 10^{23} \text{ kg}$	(A1)
	correct conclusion based on <u>comparison</u> of M with $7.4 \times 10^{22} \text{ kg}$	(B1)

33.

1(a)(i)	work done per unit mass	B1
	work done moving mass from infinity (to the point)	B1
1(a)(ii)	(near Earth's surface change in) height $\ll$ radius or height <u>much</u> less than radius	B1
	potential inversely proportional to radius and radius approximately constant (so potential approximately constant)	B1
1(b)	initial kinetic energy = (–) potential energy (at surface) or $\frac{1}{2}mv^2 = GMm/r$	B1
	$v^2 = (2 \times 6.67 \times 10^{-11} \times 7.4 \times 10^{22}) / (0.5 \times 3.5 \times 10^6)$	C1
	$v = 2.4 \times 10^3 \text{ ms}^{-1}$	A1

34.

1(a)(i)	force per unit mass	B1
1(a)(ii)	acceleration = F/m , field strength = F/m , so equal	B1
1(b)	smooth curve between R and $4R$ with negative gradient of decreasing magnitude	B1
	line passing through $(R, 1.00g)$ and $(2R, 0.25g)$	B1
	line ending at $(4R, 0.0625g)$	B1
1(c)	$M = (4/3 \times \pi R^3)\rho$	C1
	$g = GM/(2R)^2$	C1
	$g = \frac{1}{3} \times 6.67 \times 10^{-11} \times \pi \times 3.4 \times 10^6 \times 4.0 \times 10^3$ $= 0.95 \text{ ms}^{-2}$	A1

35.

1(a)(i)	work done per unit mass	B1
	idea of work done moving mass from infinity (to the point)	B1
1(a)(ii)	(gravitational) force is attractive	B1
	(gravitational) potential at infinity is zero	B1
	decrease in potential energy as masses approach or displacement and force in opposite directions	B1
1(b)(i)	Either $mv^2/R = GMm/R^2$ Or $v = \sqrt{GM/R}$ $v^2 = (6.67 \times 10^{-11} \times 6.00 \times 10^{24}) / (7.30 \times 10^6)$	C1
	giving $v = 7.4 \times 10^3 \text{ m s}^{-1}$	A1
1(b)(ii)	$V_P = -GMm/R$	C1
	$= -(6.67 \times 10^{-11} \times 6.00 \times 10^{24} \times 340) / (7.30 \times 10^6)$	C1
	$V_P = -1.9 \times 10^{10} \text{ J}$	A1
1(c)	$v^2 \propto 1/r$, (r smaller) so v greater	M1
	and E_K greater	A1

36.

1(a)	$(F =) GMm/x^2$, where G is the (universal) gravitational constant	B1
1(b)(i)	$GMm/x^2 = mv^2/x$ or $v^2 = GM/x$	C1
	$v^2 = (6.67 \times 10^{-11} \times 7.5 \times 10^{23}) / (3.4 \times 10^6 + 240 \times 10^3)$ so $v = 3.7 \times 10^3 \text{ m s}^{-1}$	A1
1(b)(ii)	potential energy $= (-)GMm/x$	C1
	$E_A = -(6.67 \times 10^{-11} \times 7.5 \times 10^{23} \times 650) / (3.64 \times 10^6)$ or $E_B = -(6.67 \times 10^{-11} \times 7.5 \times 10^{23} \times 650) / (5.00 \times 10^7)$	M1
	correct substitution and subtraction $E_B - E_A$ shown, leading to $\Delta E_p = 8.3 \times 10^9 \text{ J}$	A1
	or	
	$\phi = (-)GM/x$ and potential energy $= m\phi$	(C1)
	$\Delta\phi = (6.67 \times 10^{-11} \times 7.5 \times 10^{23}) \times [(1/(3.64 \times 10^6)) - (1/(5.00 \times 10^7))]$ ($= 1.27 \times 10^7 \text{ J kg}^{-1}$)	(M1)
	$\Delta E_p = 1.27 \times 10^7 \times 650$ $= 8.3 \times 10^9 \text{ J}$	(A1)
1(c)	kinetic energy <u>or</u> potential energy decreases	B1
	kinetic energy <u>and</u> potential energy decrease so total energy decreases	B1

37.

1(a)	$(F =) G M m / x^2$, where G is the (universal) gravitational constant	B1
1(b)(i)	angle = $(1.2 \times 10^{-3}) / (8.0 \times 10^{-2}) = 1.5 \times 10^{-2}$ (rad)	B1
1(b)(ii)	torque = $1.5 \times 10^{-2} \times 9.3 \times 10^{-10}$ = 1.4×10^{-11} N m	A1
1(c)(i)	force $\times 8.0 \times 10^{-2} = 1.4 \times 10^{-11}$	C1
	$(G \times 1.3 \times 7.5 \times 10^{-3} \times 8.0 \times 10^{-2}) / (6.0 \times 10^{-2})^2 = 1.4 \times 10^{-11}$	C1
	$G = 6.4 \times 10^{-11}$ N m ² kg ⁻²	A1
1(c)(ii)	Any one from: - law applies only to point masses/spheres are not point masses - radii of spheres not small compared with separation - spheres may not be uniform - the masses are not isolated - force between L and rod - spheres may be charged/may be electrostatic force (between spheres)	B1

38.

1(a)	force proportional to product of masses and inversely proportional to square of separation	B1
	idea of (gravitational) force between point masses	B1
1(b)(i)	above the equator	B1
	from west to east	B1
1(b)(ii)	gravitational force provides/is the centripetal force	B1
	$GM / r^2 = r (2\pi / T)^2$	C1
	$(6.67 \times 10^{-11} \times M) = \{(4.23 \times 10^7)^3 \times 4\pi^2\} / (24 \times 3600)^2$	C1
	$M = 6.0 \times 10^{24}$ kg	A1

39.

1(a)	force proportional to product of masses and inversely proportional to square of separation	B1
	idea of (gravitational) force between point masses	B1
1(b)	gravitational force provides/is the centripetal force	B1
	$GM / R^2 = R\omega^2$ or $GM / R^2 = v^2 / R$	M1
	$\omega = 2\pi / T$ or $v = 2\pi R / T$	M1
	algebra leading to $R^3 / T^2 = GM / 4\pi^2$	A1
1(c)	$(6.67 \times 10^{-11} \times M) / 4\pi^2 = (4.38 \times 10^6)^3 / (2.44 \times 3600)^2$	C1
	$M = 6.45 \times 10^{23}$ kg	A1

40.

1(a)	work done per unit mass	B1
	work done moving mass from infinity (to the point)	B1
1(b)(i)	gravitational force provides centripetal force	C1
	$mv^2/r = GMm/r^2$ and $v = 2\pi r/T$ OR $m\omega^2 r = GMm/r^2$ and $\omega = 2\pi/T$ OR $r^3 = GMT^2/4\pi^2$	C1
	$r^3 = 6.67 \times 10^{-11} \times 6.0 \times 10^{24} \times (13.7 \times 24 \times 3600)^2 / 4\pi^2$ so $r = 2.4 \times 10^8$ m	A1
1(b)(ii)	$(E_p = -) GMm/r$ work done = $GMm/r_1 - GMm/r_2$	C1
	$= 6.67 \times 10^{-11} \times 360 \times 6.0 \times 10^{24} (1/6.4 \times 10^6 - 1/2.4 \times 10^8)$	C1
	$= 2.2 \times 10^{10}$ J	A1
1(b)(iii)	$g = GM/r^2$	C1
	ratio = $r_{\text{TESS}}^2 / r_{\text{earth}}^2$ $= (2.4 \times 10^8 / 6.4 \times 10^6)^2$ $= 1400$	A1

41.

1(a)	work done per unit mass	B1
	(work done to) move mass from infinity (to the point)	B1
1(b)	curve from r to $4r$, with gradient of decreasing magnitude and starting at $(r, \pm\phi)$	B1
	line passing through $(2r, \pm 0.5\phi)$ and $(4r, \pm 0.25\phi)$	B1
	line showing potential is negative throughout	B1
1(c)(i)	gravitational potential energy = $(-) GMm/R$	C1
	change = $(6.67 \times 10^{-11} \times 6.0 \times 10^{24} \times 3.4 \times 10^3) / (6.4 \times 10^6) \times [1/3 - 1/4]$	C1
	$= 1.8 \times 10^{10}$ J	A1
1(c)(ii)	rock loses potential energy	B1
	(so) kinetic energy increases so speed increases	B1
	or	
	force is attractive	(B1)
	moves towards planet so speeds up	(B1)

42.

1(a)(i)	region (of space)	B1
	where a particle experiences a force	B1
1(a)(ii)	force per unit mass	B1
1(b)	$g = GM/R^2$	C1
	$= (6.67 \times 10^{-11} \times 6.42 \times 10^{23}) / (3.39 \times 10^6)^2$	C1
	$= 3.73$ N kg ⁻¹	A1

1(c)	$0.99 \times 3.73 = (6.67 \times 10^{-11} \times 6.42 \times 10^{23}) / r^2$	C1
	$r = 3.41 \times 10^6 \text{ (m)}$	C1
	height = $(r - R)$ $= 2 \times 10^4 \text{ m}$	A1
	or	
	$0.99 \times 3.73 = (6.67 \times 10^{-11} \times 6.42 \times 10^{23}) / (R + h)^2$ $(R + h)^2 = 1.1596 \times 10^{13}$	(C1)
	$R + h = 3.41 \times 10^6 \text{ (m)}$	(C1)
	$h = 2 \times 10^4 \text{ m}$	(A1)
	or	
	$0.99 = (3.39 \times 10^6)^2 / r^2$	(C1)
	$r = 3.41 \times 10^6 \text{ (m)}$	(C1)
	height = $2 \times 10^4 \text{ m}$	(A1)

43.

1(a)	work done per unit mass	B1
	(work done) moving mass from infinity (to the point)	B1
1(b)(i)	gravitational potential energy = $(-)GMm / r$	C1
	$\Delta E_p = 6.67 \times 10^{-11} \times 6.0 \times 10^{24} \times 2.4 \times 10^3 \times [(6.4 \times 10^6)^{-1} - (1.2 \times 10^7)^{-1}]$	C1
	or	
	$\Delta \phi = 6.67 \times 10^{-11} \times 6.0 \times 10^{24} \times [(6.4 \times 10^6)^{-1} - (1.2 \times 10^7)^{-1}]$	(C1)
	$\Delta E_p = m\Delta\phi$	(C1)
	$\Delta E_p = 7.0 \times 10^{10} \text{ J}$	A1
1(b)(ii)	$GMm / r^2 = mv^2 / r$	C1
	$v^2 = GM / r$ $= (6.67 \times 10^{-11} \times 6.0 \times 10^{24}) / (1.2 \times 10^7)$	C1
	$v = 5800 \text{ m s}^{-1}$	A1
1(c)	any one point from: - smaller gain in energy required if orbit is west to east - smaller change in velocity if orbit is west to east - smaller gain in energy if orbit is in same direction as Earth's rotation - smaller change in velocity if orbit is in same direction as Earth's rotation - satellite already moving west to east at launch - Earth's rotation is from west to east	B1

44.

1(a)	(gravitational) force is (directly) proportional to product of masses	B1
	force (between point masses) is inversely proportional to the square of their separation	B1
1(b)	correct read offs from the graph with correct power of ten for R^3	C1
	$M = \frac{4 \times \pi^2 \times 1.2 \times 10^{34}}{6.67 \times 10^{-11} \times 2.4 \times (365 \times 24 \times 3600)^2}$	C1
	$= 3.0 \times 10^{30} \text{ kg}$	A1
1(c)(i)	potential energy is zero at infinity	B1
	(gravitational) forces are attractive	B1
	work must be done on the rock to move it to infinity	B1
1(c)(ii)	$\frac{GMm}{r^2} = \frac{mv^2}{r} \quad \text{OR} \quad v^2 = \frac{GM}{r} \quad \text{OR} \quad v = \sqrt{\frac{GM}{r}}$	M1
	use of $\frac{1}{2} mv^2$ (e.g. multiplication by $\frac{1}{2} m$) leading to $\frac{GMm}{2r}$	A1
1(c)(iii)	$E_p = \phi m$ and $\phi = \frac{-GM}{r}$ or $E_p = \frac{-GMm}{r}$ Total energy = $E_k + E_p$	C1
	Total energy = $\frac{GMm}{2r} + \frac{-GMm}{r} = \frac{-GMm}{2r}$	A1

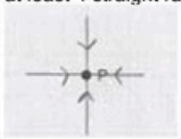
45.

1(a)	force per unit mass	B1
1(b)	$GMm/r^2 = m\omega^2$ and $\omega = 2\pi/T$ or $GMm/r^2 = mv^2/r$ and $v = 2\pi r/T$	C1
	$6.67 \times 10^{-11} \times 6.0 \times 10^{24} = r^3 \times [2\pi / (94 \times 60)]^2$	C1
	$r = 6.9 \times 10^6 \text{ m}$	A1
1(c)(i)	$r^3 \omega^2 = \text{constant}$ or $r^3/T^2 = \text{constant}$	C1
	$r^3 / (6.9 \times 10^6)^3 = (150/94)^2$ so $r = 9.4 \times 10^6 \text{ m}$	A1
	or	
	$GMT^2/4\pi^2 = r^3$ and clear that M is 6.0×10^{24}	(C1)
	$6.67 \times 10^{-11} \times 6.0 \times 10^{24} = r^3 \times [2\pi / (150 \times 60)]^2$ so $r = 9.4 \times 10^6 \text{ m}$	(A1)
1(c)(ii)	separation increases so (potential energy) increases or movement is against gravitational force so (potential energy) increases	B1
1(c)(iii)	potential energy = $(-GMm/r)$	C1
	$\Delta E_p = 6.67 \times 10^{-11} \times 6.0 \times 10^{24} \times 1200 \times [(6.9 \times 10^6)^{-1} - (9.4 \times 10^6)^{-1}]$	C1
	$= 1.9 \times 10^{10} \text{ J}$	A1

46.

1(a)	(gravitational) force per unit mass	B1
1(b)(i)	$g = GM / r^2$	C1
	$= (6.67 \times 10^{-11} \times 6.42 \times 10^{23}) / (3.39 \times 10^6)^2$ $= 3.73 \text{ N kg}^{-1}$	A1
1(b)(ii)	$a = r\omega^2$ and $\omega = 2\pi / T$ or $a = v^2 / r$ and $v = 2\pi r / T$	C1
	$a = 3.39 \times 10^6 \times (2\pi / (24.6 \times 3600))^2$ $= 0.0171 \text{ m s}^{-2}$	A1
1(b)(iii)	force per unit mass = $3.73 - 0.0171$ $= 3.71 \text{ N kg}^{-1}$	A1

47.

1(a)	at least 4 straight radial lines to P 	B1
	all arrows pointing along the lines towards P	B1
1(b)	Any 2 from: gravitational force provides the centripetal force (centripetal or gravitational) force has constant magnitude (centripetal or gravitational) force is perpendicular to velocity (of moon) / direction of motion (of moon)	B2
1(c)(i)	$\frac{GMm}{r^2} = m\omega^2$	M1
	$M = \frac{r^3\omega^2}{G}$ and gradient = $r^3\omega^2$ hence $M = \frac{\text{gradient}}{G}$ or $r^3 = GM \times 1/\omega^2$ so gradient = GM hence $M = \frac{\text{gradient}}{G}$	A1
1(c)(ii)	$M = 4.1 \times 10^{23} / (6.0 \times 10^7 \times 6.67 \times 10^{-11}) = 1.0 \times 10^{26} \text{ kg}$	B1
1(c)(iii)	$\frac{GMm}{r^2} = \frac{mv^2}{r}$ $\frac{GM}{r} = v^2$	C1
	$v^2 = \frac{6.67 \times 10^{-11} \times 1.0 \times 10^{26}}{1.2 \times 10^8}$ $v^2 = 5.6 \times 10^7 \text{ m s}^{-1}$	C1
	$v = 7500 \text{ m s}^{-1}$	A1

48.

1(a)(i)	(gravitational) force is (directly) proportional to product of masses	B1
	force (between point masses) is inversely proportional to the square of their separation	B1
1(a)(ii)	$g = F / m$	C1
	$F = GMm / r^2$	A1
	and so	
	$g = [GMm / r^2] / m = GM / r^2$	
1(b)(i)	$g = (6.67 \times 10^{-11} \times 7.35 \times 10^{22}) / (1.74 \times 10^6)^2 = 1.62 \text{ N kg}^{-1}$	A1
1(b)(ii)	fields (due to Earth and the Moon) have equal magnitudes	B1
	fields (due to Earth and the Moon) are in opposite directions	B1
1(b)(iii)	distance of X from Earth = $(3.84 \times 10^8 - x)$	C1
	$(G \times) 7.35 \times 10^{22} / x^2 = (G \times) 5.98 \times 10^{24} / (3.84 \times 10^8 - x)^2$	C1
	$x = 3.8 \times 10^7 \text{ m}$	A1

49.

1(a)(i)	work (done) per unit mass	B1
	work (done on mass) in moving mass from infinity (to the point)	B1
1(a)(ii)	$E_p = \phi m$	B1
	$E_p = (-GM / r) \times m = -GMm / r$	
	or	
	$\phi = -GM / r$ and $E_p = \phi m = -GMm / r$	
1(b)(i)	$\Delta E_p = 6.67 \times 10^{-11} \times 1.99 \times 10^{30} \times 2.20 \times 10^{14} \times [1 / (6.38 \times 10^{10}) - 1 / (8.44 \times 10^{11})]$	C1
	$= 4.23 \times 10^{23} \text{ J}$	A1
1(b)(ii)	(gravitational) force is attractive so decrease or (gravitational) force does work so decrease	B1
1(b)(iii)	$\Delta E_p = \frac{1}{2} m (v_2^2 - v_1^2)$	C1
	$4.23 \times 10^{23} = \frac{1}{2} \times 2.20 \times 10^{14} \times (v^2 - 34100^2)$	C1
	$v (= 70800 \text{ m s}^{-1}) = 70.8 \text{ km s}^{-1}$	A1
1(c)	both PE and KE equations include m , so path is unchanged	B1

50.

1(a)	$F = (Gm_1m_2) / r^2$	M1
	where G is the gravitational constant	A1
1(b)	gravitational force provides the centripetal force	B1
	$mR\omega^2 = GMm / R^2$ and $\omega = 2\pi / T$ or $mv^2 / R = GMm / R^2$ and $v = 2\pi R / T$ or $4\pi^2 mR / T^2 = GMm / R^2$	M1
	correct completion of algebra to get $T^2 = (4\pi^2 / GM) R^3$, with identification of $(4\pi^2 / GM)$ as k	A1
1(c)(i)	$(24 \times 3600)^2 = (4\pi^2 \times R^3) / (6.67 \times 10^{-11} \times 6.0 \times 10^{24})$	C1
	$R = 4.2 \times 10^7 \text{ m}$	A1
1(c)(ii)	(orbit) must be above the Equator	B1
	(direction) must be from west to east	B1

51.

1(a)	force per unit mass	B1
1(b)(i)	lines drawn are radial from the surface	B1
	arrows show pointing towards planet	B1
1(b)(ii)	field lines show force (on satellite) is towards centre of planet or velocity of satellite is perpendicular to field lines	B1
	(gravitational) force perpendicular to velocity causes centripetal <u>acceleration</u>	B1
1(c)(i)	$T = 24 \text{ hours}$	C1
	$a = r\omega^2$ and $\omega = 2\pi / T$ or $a = v^2 / r$ and $v = 2\pi r / T$ or $a = 4\pi^2 r / T^2$	C1
	$a = (4\pi^2 \times 6.4 \times 10^6) / (24 \times 60 \times 60)^2$ $= 0.034 \text{ m s}^{-2}$	A1
1(c)(ii)	identification of the two forces acting on the object as gravitational force and (normal) contact force	M1
	gravitational force and normal contact force are in opposite directions, and their resultant causes the (centripetal) acceleration	A1

52.

1(a)	work done per unit mass	B1
	work (done on mass) moving mass from infinity (to the point)	B1
1(b)(i)	$-3.55 \times 10^7 \text{ J kg}^{-1}$	B1
1(b)(ii)	$\phi = -\frac{GM}{r}$ $M = -\frac{-3.55 \times 10^7 \times 4800000}{6.67 \times 10^{-11}}$ $= 2.55 \times 10^{24} \text{ kg}$	B1
1(b)(iii)	$g = \frac{GM}{r^2}$ or $g = -\frac{\phi}{r}$	C1
	$= \frac{6.67 \times 10^{-11} \times 2.55 \times 10^{24}}{4800000^2}$ or $= \frac{3.55 \times 10^7}{4800000}$ $= 7.4 \text{ N kg}^{-1}$	A1
1(b)(iv)	r in range 2.60×10^7 to $2.65 \times 10^7 \text{ m}$	C1
	$\frac{mv^2}{r} = \frac{GMm}{r^2}$ and $v = \frac{2\pi r}{T}$ or $mr\omega^2 = \frac{GMm}{r^2}$ and $\omega = \frac{2\pi}{T}$	C1
	$T^2 = \frac{4\pi^2 r^3}{GM} = \frac{4\pi^2 \times (2.65 \times 10^7)^3}{6.67 \times 10^{-11} \times 2.55 \times 10^{24}} = 4.20 \times 10^6$	C1
	$T = 64800 \text{ s}$ $= 18 \text{ hours}$	A1
1(c)	similarity – any one point from - inversely proportional to distance (from point) - points of equal potential lie on concentric spheres - zero at infinite distance	B1
	difference – any one point from - gravitational potential is (always) negative - electric potential can be positive or negative	B1

53.

1(a)(i)	force per unit mass	B1
1(a)(ii)	force per unit positive charge	B1
1(a)(iii)	similarity: • inversely proportional to distance (from point) • points of equal potential lie on concentric spheres • zero at infinite distance <i>Any point, 1 mark</i>	B1
	difference: • gravitational potential is (always) negative • electric potential can be positive or negative <i>Any point, 1 mark</i>	B1
1(b)(i)	$g = GM / r^2$	M1
	$E = Q / 4\pi\epsilon_0 r^2$	M1
	algebra showing the elimination of r leading to $M / Q = (1 / 4\pi G \epsilon_0) (g / E)$	A1
1(b)(ii)	$\alpha = 1 / (4\pi \times 6.67 \times 10^{-11} \times 8.85 \times 10^{-12}) = 1.35 \times 10^{20} \text{ (kg}^2 \text{ C}^{-2}\text{)}$ or $\alpha = (8.99 \times 10^9) / (6.67 \times 10^{-11}) = 1.35 \times 10^{20} \text{ (kg}^2 \text{ C}^{-2}\text{)}$	A1
1(c)(i)	$E = \alpha g Q / M$	C1
	$= (1.35 \times 10^{20} \times 9.81 \times 4.80 \times 10^5) / (5.98 \times 10^{24})$	
	$= 106 \text{ NC}^{-1} \text{ or } 106 \text{ V m}^{-1}$	A1
1(c)(ii)	same (direction)	B1

54.

1(a)	(gravitational) force is (directly) proportional to product of masses	B1
	force (between point masses) is inversely proportional to the square of their separation	B1
1(b)	$GMm / R^2 = mR\omega^2$	M1
	$\omega = 2\pi / T$ <u>and</u> algebra leading to $4\pi^2 R^3 = GMT^2$	A1
	or	
	$GMm / R^2 = mv^2 / R$	(M1)
	$v = 2\pi R / T$ <u>and</u> algebra leading to $4\pi^2 R^3 = GMT^2$	(A1)
1(c)	$4\pi^2 \times R^3 = 6.67 \times 10^{-11} \times 5.98 \times 10^{24} \times (24 \times 60 \times 60)^2$	C1
	$(R = 4.22 \times 10^7 \text{ m})$	
	$h = R - (6.37 \times 10^6)$	C1
	$h = (4.22 \times 10^7) - (6.37 \times 10^6)$ $= 3.6 \times 10^7 \text{ m}$	A1
1(d)(i)	$\omega = 2\pi / T$	C1
	$= 2\pi / (24 \times 60 \times 60)$	A1
	$= 7.3 \times 10^{-5} \text{ rad s}^{-1}$	
1(d)(ii)	orbit is from east to west	B1
	orbit is not equatorial / orbit is polar	B1

55.

1(a)(i)	direction of the force acting on a (test) mass placed at the point	B1
1(a)(ii)	change in height negligible compared with radius (of Earth)	B1
	(so) field lines are (effectively) parallel	B1
1(b)(i)	$Y = GM/R^2$	M1
	G is the gravitational constant	A1
1(b)(ii)	gravitational force is (always) attractive or gravitational force (always) acts towards the centre of the sphere	B1
	force is in opposite direction to displacement or at a point to the right of the centre, force acts to the left or at a point to the left of the centre, force acts to the right	B1
1(b)(iii)	sketch: smooth curve with decreasing positive gradient, starting at $(R, -Y)$ and reaching $3R$ with g still negative or smooth curve with increasing positive gradient, ending at $(-R, Y)$ and reaching $-3R$ with g still positive	B1
	both of the above curves, in correct quadrants	B1
	curve passing through $(\pm 2R, \pm 0.25 Y)$ and $(\pm 3R, \pm 0.11 Y)$	B1